

Constructing a **tamper proof** synthesis log for DNA benchtop devices



How can we prevent benchtop operators from modifying the sequences they synthesise to hide suspicious behaviour?

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Policy landscape

Oct 2023 **HHS Screening Framework Guidance** — brings benchtop manufacturers into scope; recommends an on-device data-logging function.

Apr 2024 **OSTP Framework for Nucleic Acid Synthesis Screening** — screening records retained three years. Currently under revision.

2024 **UK Screening Guidance** — baseline expectations covering benchtop equipment, at a 50-nucleotide threshold.

Threat model

The operator owns the device: they run it, hold root, and can reflash its firmware. The design treats them as the adversary — one who has synthesised a sequence of concern and wants the record to look clean.

ACCESS — HOW MUCH REACH THE ATTACKER HAS

REMOTE, LIMITED Brief network access via a time-boxed exploit.	REMOTE, PERSISTENT Undetected, ongoing access to the synthesiser's network.	PHYSICAL, LIMITED Hands-on for certain hours — e.g. a lab out of working hours.	PHYSICAL, PERSISTENT Owns the machine: continuous, unlimited possession.
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SKILL — HOW CAPABLE THE ATTACKER IS

NOVICE Minimal skill; manual tampering.	COMPETENT Scripts and public CVEs. < \$1k.	MODERATE Common software and basic hardware attacks. < \$10k.	ADVANCED Custom malware, side-channel work. < \$100k.
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WHAT THEY TRY TO DO

- **Edit or delete a past entry** — rewrite the record of what was synthesised.
- **Fork or truncate the log** — roll it back to an earlier state, or drop the entries made while offline.
- **Forge a signature** — extract the key, or get their own entries signed, so a tampered log still verifies.

What is a hash function?

A hash function takes an input of any size — a sequence, a log entry, an entire log — and returns a fixed-length string. Everything here rests on two properties of that string.

```
HASH(ATGCGCATTCGGATCCAA...) = 6139df0e5f71a4c8...
HASH(ATGCGCAATCGGATCCAA...) = b7c204ee2a1f882e...
HASH(the entire synthesis log) = a4dd7f310c8a4b6e...
```

Pseudorandom. There is no correlation between the function's input and its output, so that there is no way to predict outputs.

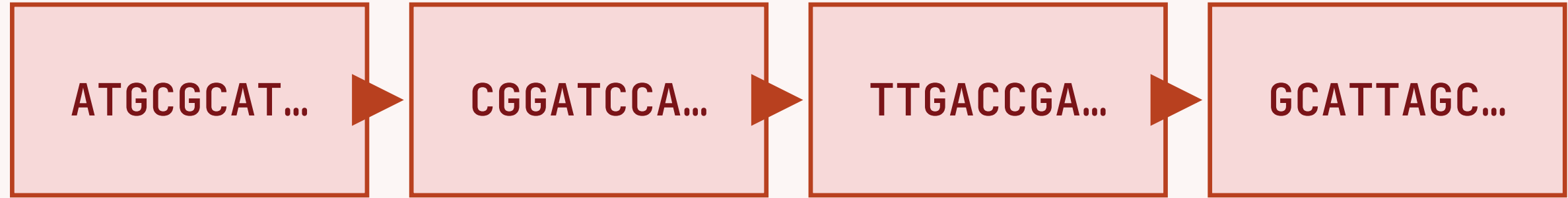
Effectively one-to-one. For SHA-256 no two distinct inputs have ever been found to give the same output. In practice the hash identifies its input uniquely, and that is what lets a single value stand in for an entire log.

Hash chain vs Merkle tree - Design decision

Both detect tampering. But merkle trees prevent disclosing sequence information to an auditor checking for tampering whereas hash chains do.

HASH CHAIN

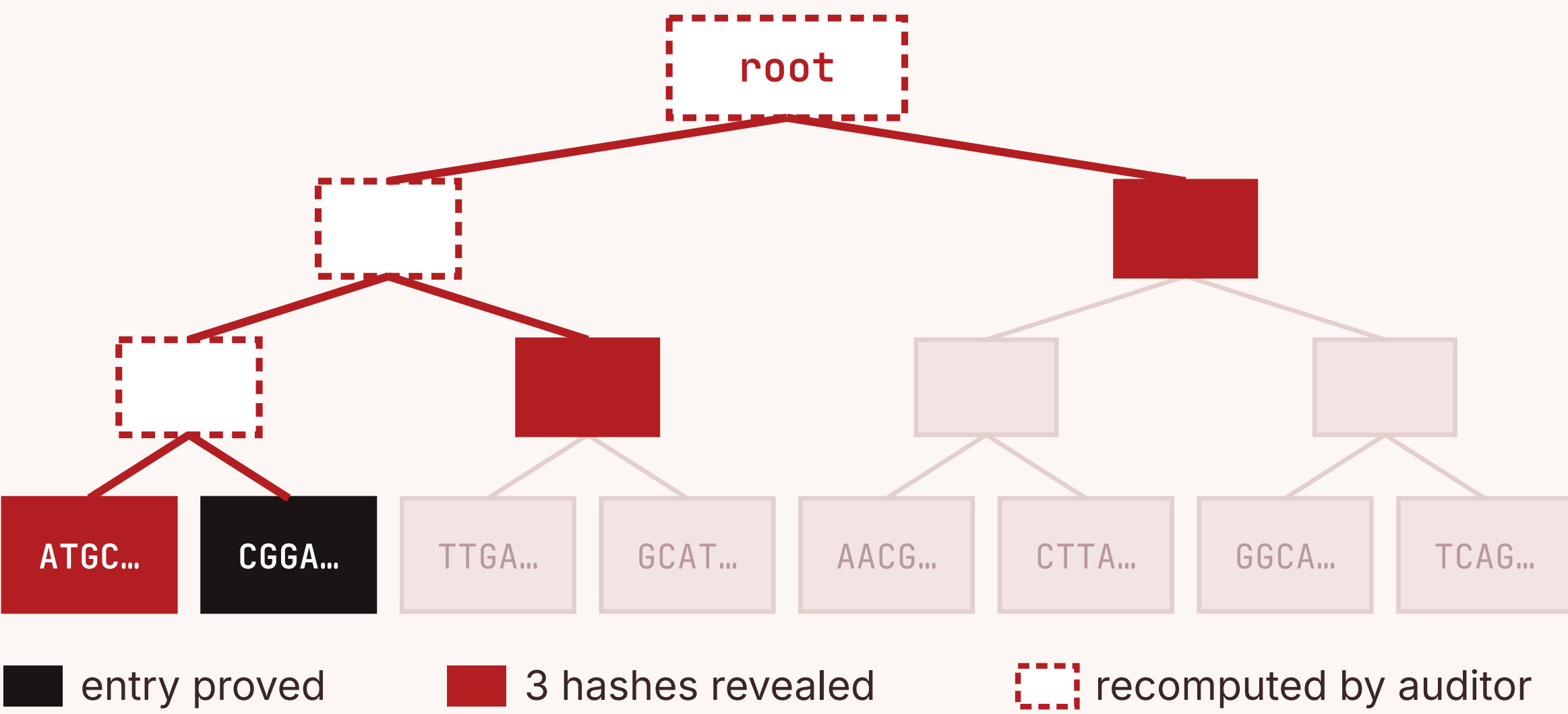
To verify one entry you must walk the chain — every entry is revealed.



Every sequence in the log is handed to the auditor

MERKLE TREE

One entry's inclusion is proved with a short list of sibling hashes.



■ entry proved ■ 3 hashes revealed □ recomputed by auditor

Proving entry d₂ was logged. The chain must be walked end to end, so every sequence is disclosed; the tree needs only three sibling hashes, from which no sequence can be recovered.

1,000

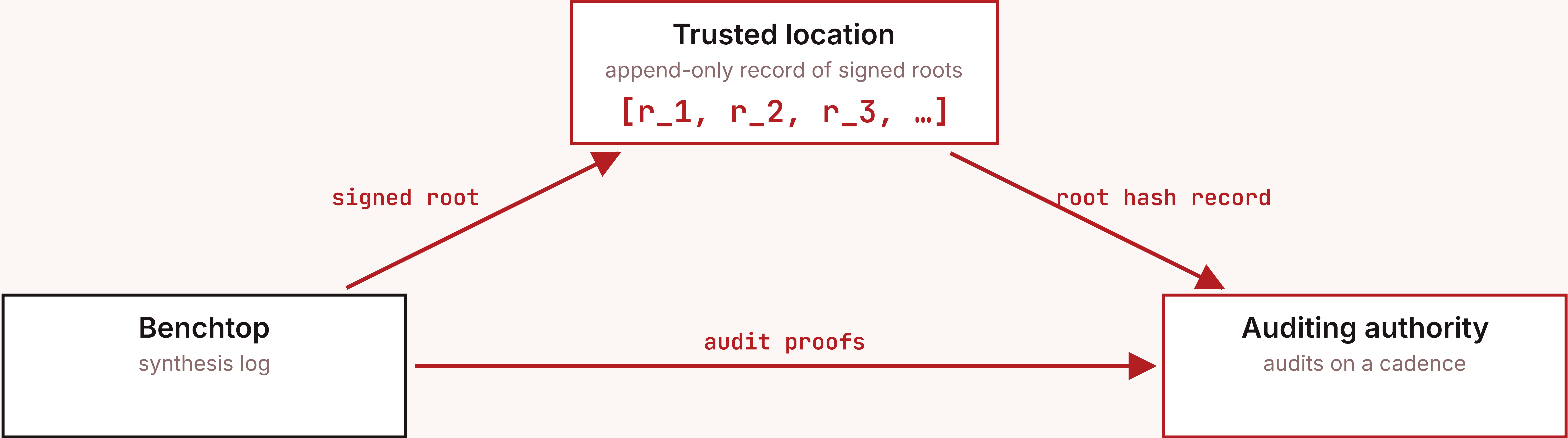
entries an auditor reads to verify one synthesis in a 1,000-entry log, using a hash chain.

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hashes for the same check with a Merkle tree — log₂(n), and not one of them a sequence.

Recommend: Auditing authority + Store cryptographic root hashes off-device

The design needs a party that is not the operator, holding the record of root hashes and the decryption key and performing the audits. No such body exists for benchtop synthesis records today.



The auditor compares the log the operator hands over against roots the operator never controlled. Tampering shows up as a mismatch between the two arrows arriving on the right.

Natural candidates already hold relationships with benchtop manufacturers and synthesis providers — **IBBIS**, which has run test screens against benchtop devices, or **SecureDNA**. The cryptography is the easy half; the institution is what has to be built.